PS#10 - Separable differential equations - Answer key

1. (AC 7.4 Exercise 6) The mass of a radioactive sample decays at a rate that is proportional to its mass.

   (a) Express this fact as a differential equation for the mass $M(t)$ using k for the constant of proportionality.

   $$\frac{dM}{dt} = -kM$$

   (Setting it up this way, with a negative sign in front of kM, assumes that $k > 0$ and tricks my brain into believing that this is indeed decay rather than growth.)

   (b) If the initial mass is M_0, find an expression for the mass $M(t)$.

   Separate and integrate. I'm going to use the usual trick of "multiplying by dt" which, we remember, is not exactly what's happening. I'm also going to use the usual trick of relabeling e^C as a new constant to deal with the $\pm$ issue that would arise from dropping the absolute values on M.

   $$\begin{aligned}
   \frac{dM}{dt} &= -kM \\
   \frac{1}{M}\,dM &= -k\,dt \\
   \int \frac{1}{M}\,dM &= \int -k\,dt \\
   \ln|M| &= -kt + C \\
   |M| &= e^{-kt+C} = e^{-kt}e^C \\
   M(t) &= Ce^{-kt} \\
   M_0 = M(0) &= Ce^0 = C \\
   M(t) &= M_0 e^{-kt}
   \end{aligned}$$

   (c) The *half-life* of the sample is the amount of time required for half of the mass to decay. Knowing that the half-life of Carbon-14 is 5730 years, find the value of k for a sample of Carbon-14.

   From this information I know that $M(5730) = \frac{1}{2}M_0$. Therefore,

   $$\begin{aligned}
   M_0 e^{-k\cdot 5730} &= \frac{1}{2}M_0 \\
   -k\cdot 5730 &= \ln\left(\frac{1}{2}\right) = -\ln(2) \\
   k &= \frac{\ln(2)}{5730}
   \end{aligned}$$

   (So: in general, for a radioisotope with a half-life of h years, the value of k works out to be $\frac{\ln(2)}{h}$.)

(d) How long does it take for a sample of Carbon-14 to be reduced to one-quarter its original mass?

We want to solve $M(t) = \frac{1}{4}M_0$. Here comes some logarithm algebra:

$$\begin{aligned} M_0 e^{-kt} &= \frac{1}{4}M_0 \\ -kt &= \ln\left(\frac{1}{4}\right) = -\ln(4) \\ t &= \frac{\ln(4)}{k} = \frac{\ln(4)}{\ln(2)/5730} = 5730 \cdot \frac{\ln(4)}{\ln(2)} = 5730 \cdot \frac{\ln(2^2)}{\ln(2)} = 5730 \cdot \frac{2\ln(2)}{\ln(2)} = 11460 \end{aligned}$$

(a) logarithms are so cool; (b) this makes sense because to get to a quarter we should halve twice.

(e) Carbon-14 naturally occurs in our environment; any living organism takes in Carbon-14 when it eats and breathes. Upon dying, however, the organism no longer takes in Carbon-14. Suppose that you find remnants of a pre-historic firepit. By analyzing the charred wood in the pit, you determine that the amount of Carbon-14 is only 30% of the amount in living trees. Estimate the age of the firepit.

So here we're going to solve $M(t) = 0.3M_0$:

$$\begin{aligned} M_0 e^{-kt} &= 0.3M_0 \\ -kt &= \ln(0.3) \\ t &= \frac{\ln(0.3)}{-k} = \frac{\ln(0.3)}{-\ln(2)/5730} = 5730 \cdot \frac{\ln(0.3)}{-\ln(2)} \approx 9952.81 \end{aligned}$$

I'm going to conclude that the firepit is about ten thousand years old, give or take.

2. (AC 7.4 Exercise 7) Consider the initial value problem

$$\frac{dy}{dt} = -\frac{t}{y},\ y(0) = 8$$

(a) Find the solution of the initial value problem and sketch its graph.

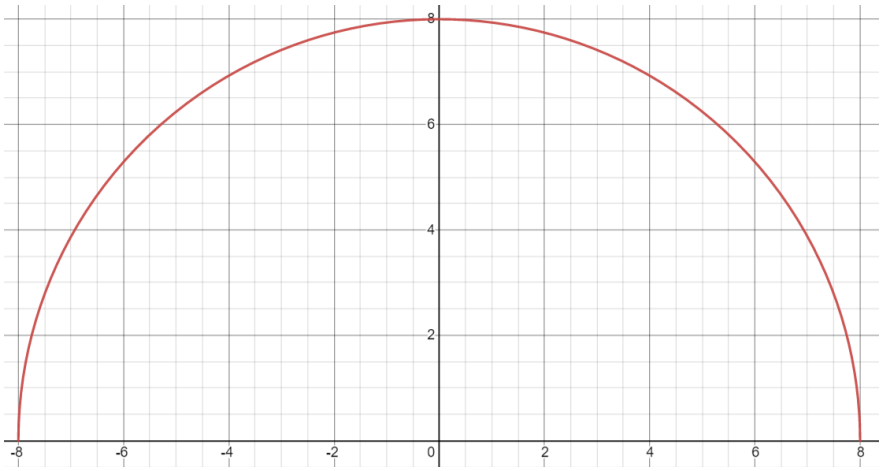
I'm going to use the usual trick of "multiplying both sides by dt" even though I know that's not really what's happening:

$$\begin{aligned} \frac{dy}{dt} &= -\frac{t}{y} \\ y\,dy &= -t\,dt \\ \int y\,dy &= \int -t\,dt \\ \frac{y^2}{2} &= \frac{-t^2}{2} + C \\ y^2 &= C - t^2 \end{aligned}$$

I will pause here to note that $t^2 + y^2 = C$ is totally the equation of a circle, but that's not a function. Let's figure out the solution to this IVP; we're going to need to take a square root, and the initial value will help us decide about the $\pm$:

$$y(t) = \pm\sqrt{C - t^2}$$
$$8 = y(0) = \pm\sqrt{C - 0^2}$$

This means that we'll have to take the $+$ and that $C = 64$. Therefore, our solution is $y(t) = \sqrt{64 - t^2}$ – the top half of a circle with radius 8.



(b) For what values of t is the solution defined?

Since the square root of a negative number isn't defined (in the real numbers, anyway), $y(t)$ is only defined for t in the range $-8 \leq t \leq 8$.

(c) What is the value of y at the last time that the solution is defined?

When $t = 8$, $y(8) = \sqrt{64 - 8^2} = 0$.

(d) By looking at the differential equation, explain why we should not expect to find solutions with the value of y you noted in part (c).

Yeah, so that slope is wild: $\frac{dy}{dt} = -\frac{t}{y} = -\frac{8}{0}$. (Indeed, the tangent line to the side of the circle is vertical and thus has an undefined slope.)

3. (AC 7.4 Exercise 8) Suppose that a cylindrical water tank with a hole in the bottom is filled with water. The water, of course, will leak out and the height of the water will decrease. Let $h(t)$ denote the height of the water. A physical principle called *Torricelli's Law* implies that the height decreases at a rate proportional to the square root of the height.

(a) Express this fact using k as the constant of proportionality.

$$\frac{dh}{dt} = k\sqrt{h}$$

(b) Suppose you have two tanks, one with $k = -1$ and another with $k = -10$. What physical differences would you expect to find?

The tank with $k = -1$ is leaking much slower than the tank with $k = -10$. I'm going to say that the tank with $k = -10$ has a way smaller hole in the bottom.

(c) Suppose you have a tank for which the height decreases at 20 inches per minute when the water is filled to a depth of 100 inches. Find the value of k.

So this information is telling us that when $h = 100$, $\dfrac{dh}{dt} = -20$. Putting this information into our differential equation, we can figure out the value of k:

$$\begin{aligned} \frac{dh}{dt} &= k\sqrt{h} \\ -20 &= k\sqrt{100} \\ -20 &= 10k \\ k &= -2 \end{aligned}$$

(d) Solve the initial value problem for the tank in part (c), and graph the solution you determine.

I'm going to use the usual trick of "multiplying both sides by dt" even though I know that's not really what's happening:

$$\begin{aligned} \frac{dh}{dt} &= -2\sqrt{h} \\ \frac{1}{\sqrt{h}}\,dh &= -2\,dt \\ \int h^{-1/2}\,dh &= \int -2\,dt \\ \frac{h^{1/2}}{1/2} &= -2t + C \\ \sqrt{h} &= -t + C \\ h(t) &= (-t + C)^2 \end{aligned}$$

To figure out the value of C, we're going to say that time $t = 0$ is the moment when the tank was filled to a depth of 100 inches. (That is, any other times we end up figuring out in this problem are measured in "minutes after the tank had 100 inches of water in it.")

$$\begin{aligned} 100 = h(0) &= (-0 + C)^2 \\ 10 &= C \\ h(t) &= (-t + 10)^2 \end{aligned}$$



(e) How long does it take for the water to run out of the tank?

$h(t) = 0$ when $t = 10$, so it takes 10 minutes (after the tank had 100 inches of water in it) for all the water to drain out.

(f) Is the solution that you found valid for all time t? If so, explain how you know this. If not, explain why not.

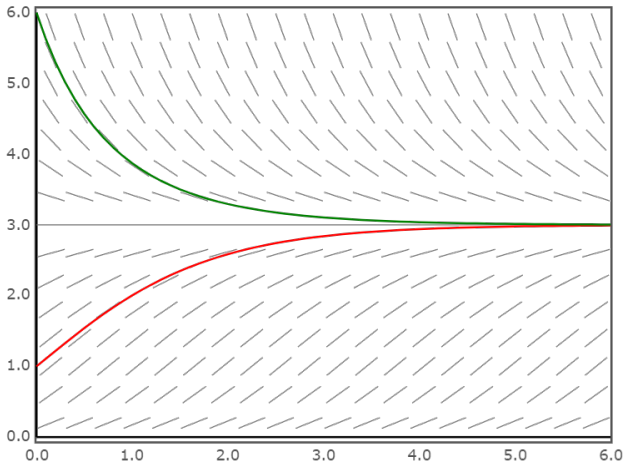
The solution predicts that the tank will start *filling* again after $t = 10$, which is clearly a silly prediction. So the solution certainly isn't valid beyond $t = 10$.

4. (AC 7.4 Exercise 9) The *Gompertz equation* is a model that is used to describe the growth of certain populations. Suppose that $P(t)$ is the population of some organism and that

$$\frac{dP}{dt} = -P\ln\left(\frac{P}{3}\right) = -P(\ln P - \ln 3)$$

(a) Sketch a slope field for $P(t)$ over the range $0 \le P \le 6$.

This also includes the solution curves at $P(0) = 1$ and $P(0) = 6$. Notice that I'm only including positive values for t.



(b) Identify any equilibrium solutions and determine whether they are stable or unstable.

Looks like there's a stable equilibrium at $P = 3$, and indeed, when we set the differential equation equal to 0 and solve for P, we get $-P\ln\left(\frac{P}{3}\right) = 0$, so either $P = 0$ or $\ln\left(\frac{P}{3}\right) = 0$. $P = 0$ isn't a valid equilibrium solution because putting $P = 0$ into the differential equation causes us to take $\ln(0)$, which doesn't exist, so we'll discard that one as extraneous. Therefore, we're left with $\ln\left(\frac{P}{3}\right) = 0$, so $\frac{P}{3} = 1$, so $P = 3$.

(c) Find the population $P(t)$ assuming $P(0) = 1$ and sketch its graph. What happens to $P(t)$ after a very long time?

I'm going to use the usual trick of "multiplying both sides by dt" even though I know that's not really what's happening. I'm also going to use the usual trick of relabeling e^C as a new constant to deal with the $\pm$ issue that would arise from dropping the absolute values.

$$\frac{dP}{dt} = -P(\ln P - \ln 3)$$

$$\frac{1}{-P(\ln P - \ln 3)}\,dP = 1\,dt$$

$$\int \frac{1}{-P(\ln P - \ln 3)}\,dP = \int 1\,dt = t + C$$

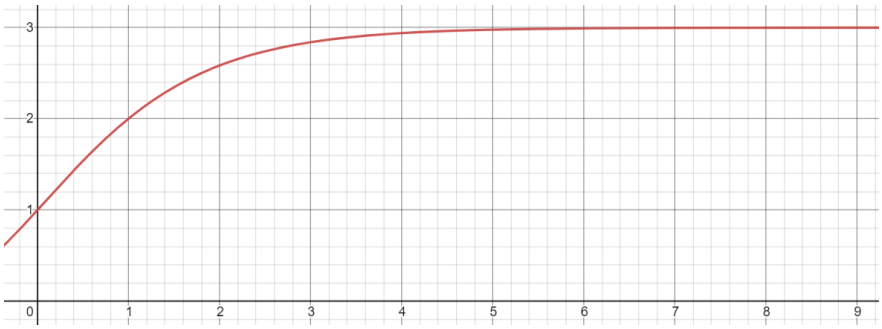
We'll make a u-substitution with $u = \ln P - \ln 3$. Then $du = \frac{1}{P}\,dP$.

$$\int \frac{1}{-u}\,du = t + C$$

$$
\begin{aligned}
-\ln|u| &= t + C, \text{ so } \ln|u| = -t + C \\
\ln|\ln P - \ln 3| &= -t + C \\
|\ln P - \ln 3| &= e^{-t+C} = e^{-t}e^{C} \\
\ln P - \ln 3 &= Ce^{-t} \\
\ln P &= Ce^{-t} + \ln 3 \\
P(t) &= e^{Ce^{-t}+\ln 3} = e^{Ce^{-t}}e^{\ln 3} = 3e^{Ce^{-t}}
\end{aligned}
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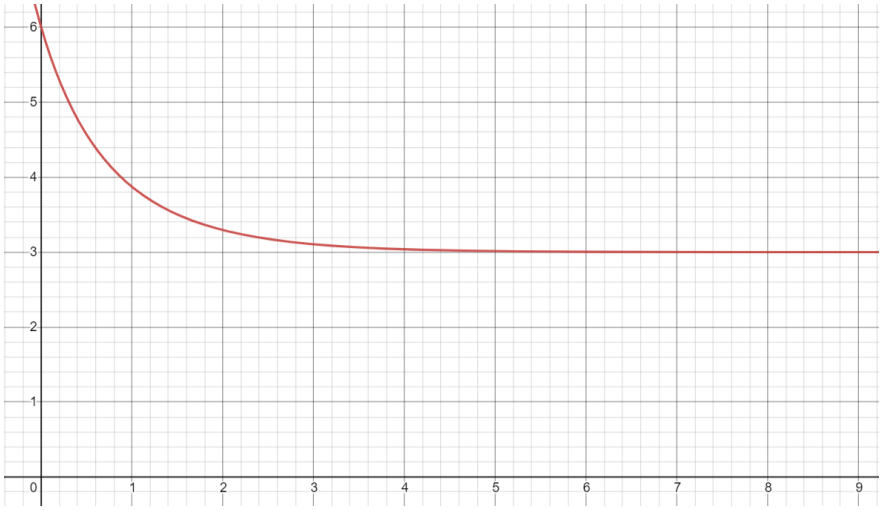
(Type $P(t) = 3e^{Ce^{-t}}$ into Desmos and play with the slider that comes up for C!)

At this point we can use our initial condition to find the value of C. If $1 = P(0) = 3e^{Ce^{-0}} = 3e^{C}$, then
$C = \ln(1/3) = -\ln 3$. Therefore, our solution is $P(t) = 3e^{-\ln(3)e^{-t}}$. It appears that the long-run behavior of this solution is indeed to tend toward our stable equilibrium at $P = 3$.



(d) Find the population $P(t)$ assuming $P(0) = 6$ and sketch its graph. What happens to $P(t)$ after a very long time?

If $6 = P(0) = 3e^{Ce^{-0}} = 3e^{C}$, then $C = \ln(2)$. Therefore, our solution is $P(t) = 3e^{\ln(2)e^{-t}}$. It appears that the long-run behavior of this solution is indeed to tend toward our stable equilibrium at $P = 3$.



(e) Verify that the long-term behavior of your solutions agrees with what you predicted by looking at the slope field.

Yep, this looks right: both of our solutions are indeed trending toward the stable equilibrium at $P = 3$.